

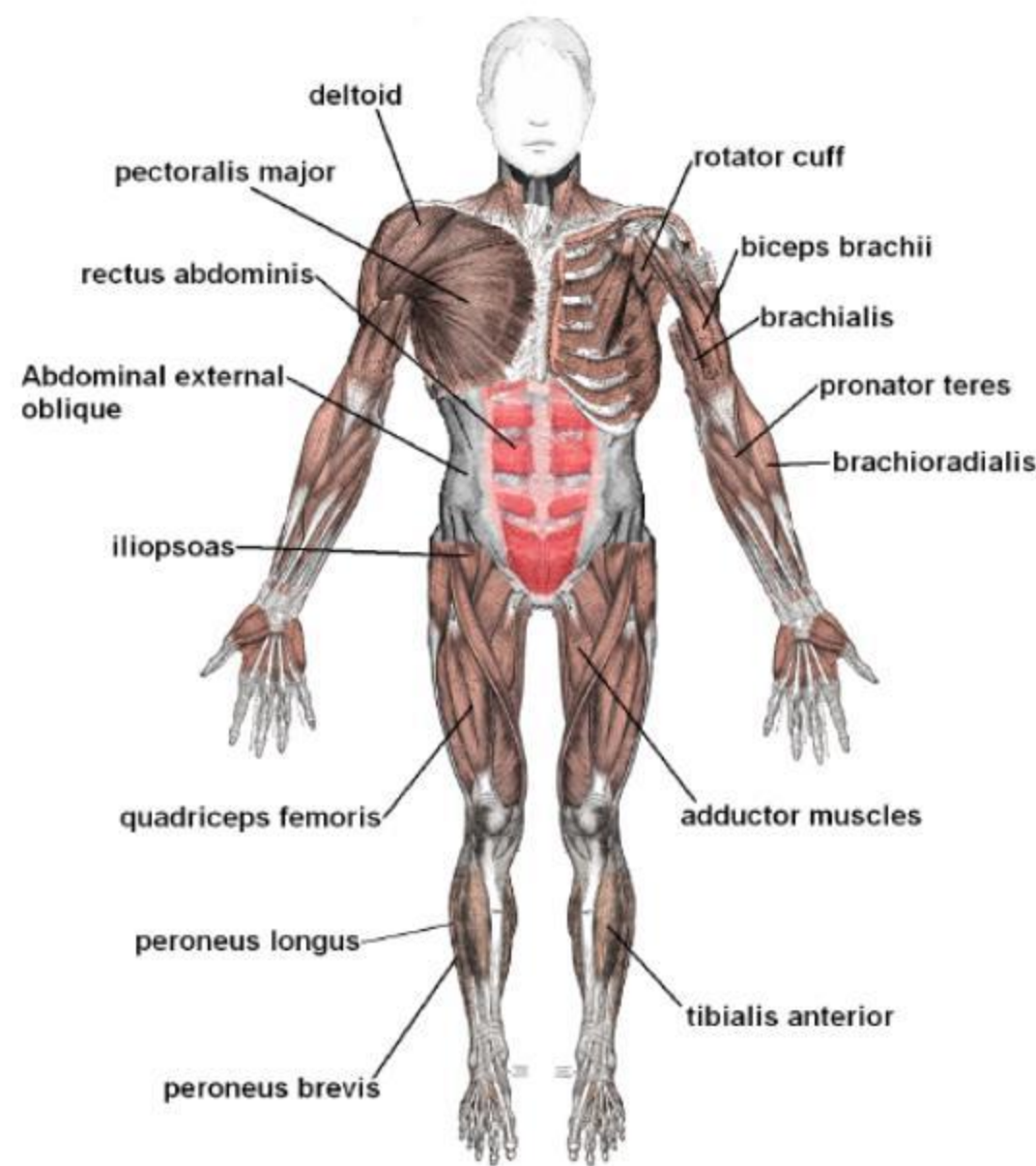


Figure 6.3.3: This figure shows major skeletal muscles in the front (anterior) of the body.

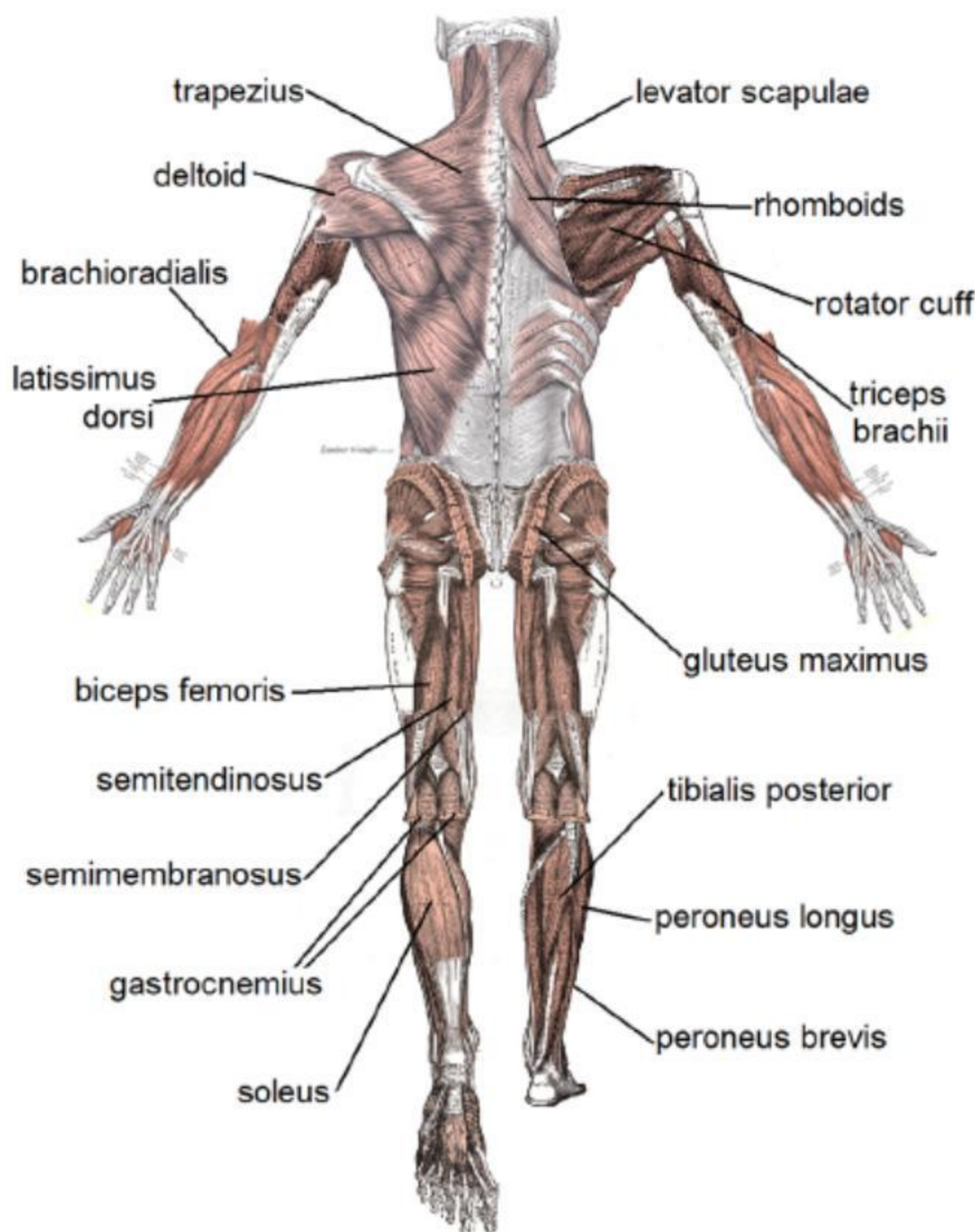


Figure 6.3.4: This figure shows major skeletal muscles in the back (posterior) of the body.

SKELETAL MUSCLE PAIRS

To move bones in opposite directions, skeletal muscles often consist of muscle pairs that work in opposition to one another. For example, when the biceps muscle (on the front of the upper arm) contracts, it can cause the elbow joint to flex or bend the arm, as shown in Figure 6.3.5. When the triceps muscle (on the back of the upper arm) contracts, it can cause the elbow to extend or straighten the arm. The

biceps and triceps muscles are examples of a muscle pair where the muscles work in opposition to each other.

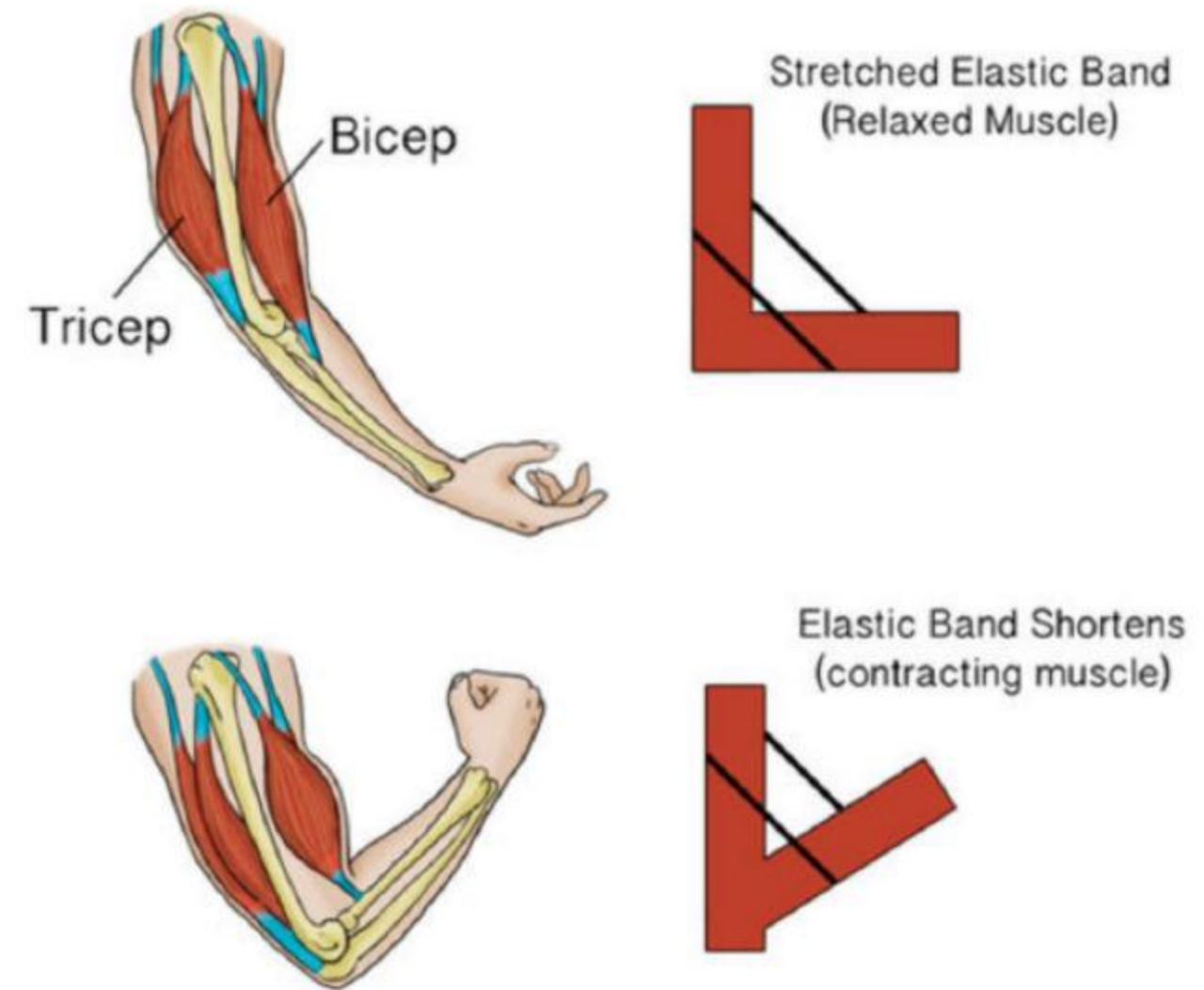


Figure 6.3.5: Triceps and biceps muscles in the upper arm are opposing muscles that move the arm at the elbow in opposite directions.

SKELETAL MUSCLE STRUCTURE

Each skeletal muscle consists of hundreds — or even *thousands* — of skeletal muscle fibers, which are long, string-like cells. As shown in Figure 6.3.6, skeletal muscle fibers are individually wrapped in connective tissue called **endomysium**. The skeletal muscle fibers are bundled together in units called **muscle fascicles**, surrounded by sheaths of connective tissue called **perimysium**. Each fascicle contains between ten and 100 (or even more!) skeletal muscle fibers. Fascicles, in turn, are bundled together to form individual skeletal muscles, which are wrapped in connective tissue called **epimysium**. The connective tissues in skeletal muscles have a variety of functions. They support and protect muscle fibers, allowing them to withstand contraction forces by distributing the forces applied to the muscle. They also provide pathways for nerves and blood vessels to reach the muscles. Also, the epimysium anchors the muscles to tendons.

Structure of a Skeletal Muscle

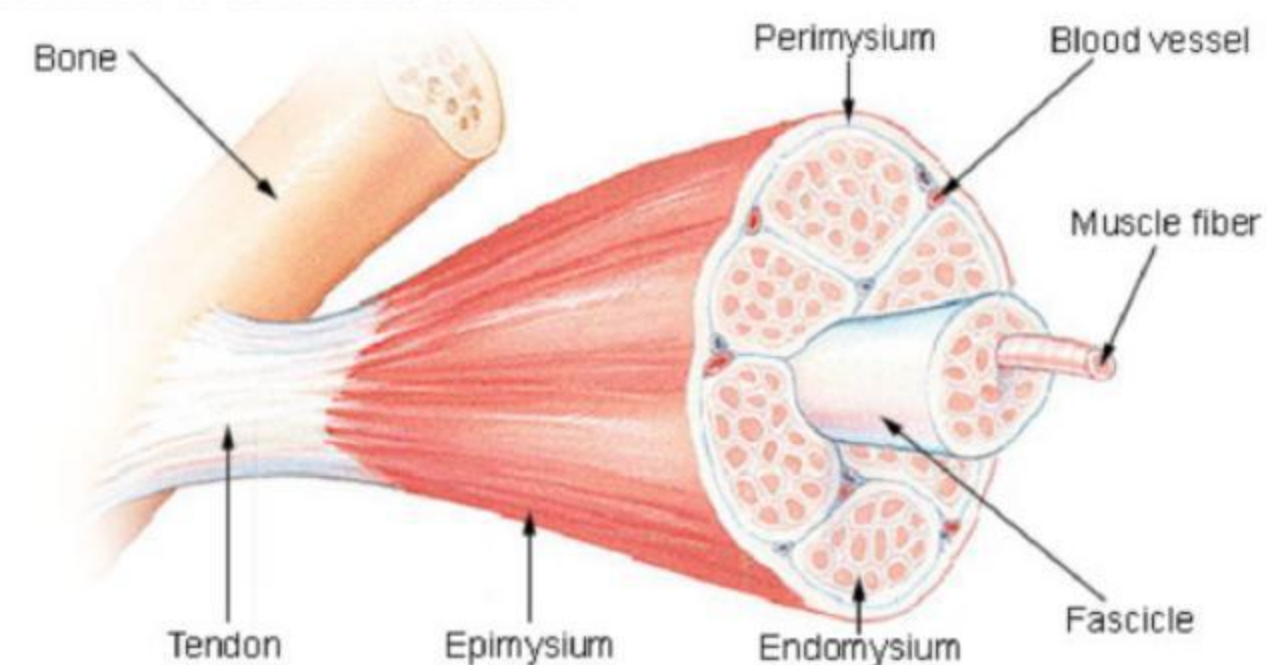


Figure 6.3.6: Each skeletal muscle has a structure of bundles within bundles. Bundles of muscle fibers make up a muscle fascicle, and fascicles' bundles make up a skeletal muscle. At each level of bundling, a connective tissue membrane surrounds the bundle. The muscle cells, fascicle, and the whole muscle are surrounded by Endomysium, perimysium, and epimysium, respectively. All connective tissues merge together to make a tendon that attaches the muscle to bones.